

Inductive bias meta-learning with generative models

Anna Novokshonova

Consultant: Fedor Sobolevsky, Muhammadsharif Nabiev

Expert: Oleg Bakhteev, PhD

Course: My first scientific paper

Intelligent Systems Department, MIPT

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Goal of research

Goal

Extraction of inductive bias of differentiable learning algorithms.

Problem

The existing solution assumes prior knowledge of the data distribution or requires a fixed dataset.

Solution

A meta-learning algorithm that trains a generator to produce datasets that are easy to generalize to the target model and that reflect its inductive bias.

Extracting inductive bias via generating easy-to-generalize datasets

Previous work^a

Meta-learning a function that assigns labels to samples from a fixed dataset or data distribution.

Our extension

Meta-learning a generative model to produce entire synthetic datasets.

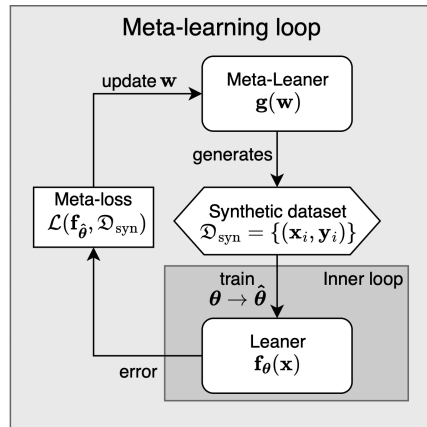


Figure: Meta-learning scheme

^aWilliam Dorrell, Maria Yuffa, and Peter Latham. *Meta-Learning the Inductive Biases of Simple Neural Circuits*. 2023. arXiv: 2211.13544 [q-bio.NC]. URL: <https://arxiv.org/abs/2211.13544>.

Problem statement

Given a target model $\mathbf{f}_{\boldsymbol{\theta}} : \mathbb{X} \rightarrow \mathbb{Y}$ with loss function $\ell(\mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}), \mathbf{y})$, the generator $\mathbf{g}(\mathbf{w})$ defines a parametric family of conditional distributions $p(\mathbf{x} \mid \mathbf{y}, \mathbf{w})$ over inputs $\mathbf{x}$ given labels $\mathbf{y}$.

Required to find data distribution function p^* , corresponding to parameters $\mathbf{w}^*$, the solution to the following optimization problem:

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \mathbb{E}_{\mathcal{D}_{\text{syn}} \sim p(\mathbf{x}|\mathbf{y}, \mathbf{w})} [\mathcal{L}(\mathbf{f}_{\hat{\boldsymbol{\theta}}}, \mathcal{D}_{\text{syn}})] \quad \text{s.t.} \quad |\mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{test}}| = d$$

where $\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{train}}} \ell(\mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}), \mathbf{y})$ and $\mathcal{L}$ is meta loss function.

Meta loss function

The meta-loss function $\mathcal{L}$ combines prediction error $\mathcal{L}_g$ and regularization $\mathcal{R}$:

$$\mathcal{L} = \mathcal{L}_g + \lambda \mathcal{R}$$

Prediction error $\mathcal{L}_g$ (MSE or cross-entropy):

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{test}}} \sum_{i=1}^m y_i \log ([\mathbf{f}_{\hat{\theta}}(\mathbf{x})]_i)$$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{test}}} \|\mathbf{y} - \mathbf{f}_{\hat{\theta}}(\mathbf{x})\|^2$$

Regularization $\mathcal{R}$:

$$\mathcal{R}_{\text{dist}} = \frac{1}{C} \sum_{k=1}^C \frac{1}{n_k(n_k - 1)} \sum_{\substack{i, j \in \mathcal{I}_k \\ i \neq j}} \|\mathbf{x}_i - \mathbf{x}_j\|_2^2, \quad \mathcal{R}_{\text{SVD}} = \frac{1}{r} \sum_{k=1}^r (\max(\tau - \sigma_k, 0))^2$$

Algorithm 1 Meta-learning for inductive bias extraction

- 1: Initialize generator $g(\mathbf{w})$ with random parameters $\mathbf{w}$
 - 2: **while** step $<$ total steps **do**
 - 3: Generate synthetic dataset $\mathcal{D}_{\text{syn}} = \{(\mathbf{x}_i, \mathbf{y}_i)\}$ with C classes and N samples per class
 - 4: Split $\mathcal{D}_{\text{syn}}$ into training set $\mathcal{D}_{\text{train}}$ and test set $\mathcal{D}_{\text{test}}$
 - 5: Re-initialize target model $\mathbf{f}_{\theta}$ from scratch
 - 6: Train $\mathbf{f}_{\theta}$ on $\mathcal{D}_{\text{train}}$ in the inner loop $\rightarrow$ trained model $\mathbf{f}_{\hat{\theta}}$
 - 7: Compute meta-loss: $\mathcal{L}_{\text{meta}} = \mathcal{L}_g(\mathbf{f}_{\hat{\theta}}, \mathcal{D}_{\text{test}}) + \lambda \mathcal{R}(\mathcal{D}_{\text{syn}})$
 - 8: Take $\mathbf{w}$ gradient step on meta-loss
 - 9: **end while**
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CNN and 1x4 images

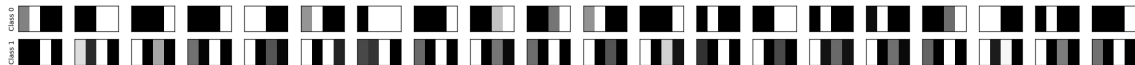


Figure: Obtained dataset for average intra-class distance regularization

Generator: Trainable tensor.

CNN Architecture: Conv1d with kernel 1×2 , stride (1,2) $\rightarrow$ max pooling $\rightarrow$ linear layer $\rightarrow$ class logits.

Training	Test	Accuracy (%)
Original	Original	98.78 ± 8.82
Original	Swapped	98.78 ± 8.82
Swapped	Swapped	99.08 ± 7.25
Swapped	Original	99.08 ± 7.25

Table: Accuracy comparison on original and swapped data

Parameters:

1. 2 classes, 20 samples per class
2. 7 steps in inner loop
3. Learning rates: inner = 0.5, outer = 0.05

Meta-loss function: $\mathcal{L} = \mathcal{L}_{MSE} + 0.5 \cdot \mathcal{R}_{dist}$

CNN and 4x4 images: locality

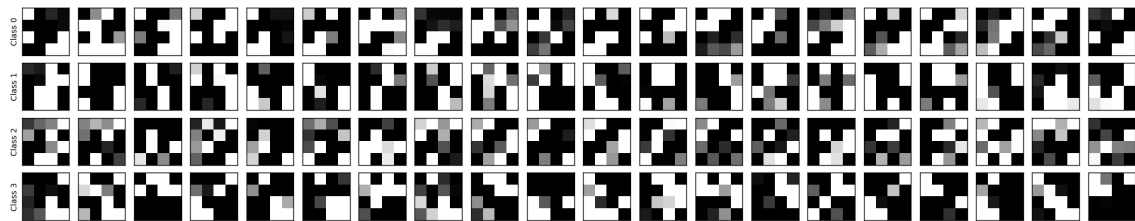


Figure: Obtained dataset for cross-entropy and average intra-class distance regularization

Generator: Trainable tensor.

CNN Architecture: Conv2d with kernel 2×2 , stride 2 $\rightarrow$ ReLU $\rightarrow$ max pooling (2×2) $\rightarrow$ flatten $\rightarrow$ linear layer $\rightarrow$ class logits.

Parameters: 4 classes, 20 samples per class, 10 steps in inner loop, learning rates of 0.5 (inner) and 0.01 (outer).

Meta-loss function: $\mathcal{L}_{CE} / \mathcal{L}_{MSE} + \mathcal{R}_{dist} / \mathcal{R}_{SVD}$ (different combinations).

CNN and 4x4 images: locality

Setup	Original	Seed 42	Seed 123	Seed 456	Seed 789
CE+Dist	98.87 ± 5.33	98.87 ± 5.33	98.87 ± 5.33	98.87 ± 5.33	98.87 ± 5.33
MSE+Dist	99.03 ± 5.22	99.03 ± 5.22	99.03 ± 5.22	99.03 ± 5.22	99.03 ± 5.22
CE+SVD	99.33 ± 4.62	99.33 ± 4.62	99.33 ± 4.62	99.33 ± 4.62	99.33 ± 4.62
MSE+SVD	98.96 ± 5.46	98.96 ± 5.46	98.96 ± 5.46	98.96 ± 5.46	98.96 ± 5.46
MLP	62.24 ± 14.73	78.76 ± 13.02	77.41 ± 12.97	77.61 ± 12.97	76.05 ± 13.68

Table: Accuracy on original and shuffled data for all configurations

CNN and 4x4 images: translation invariance

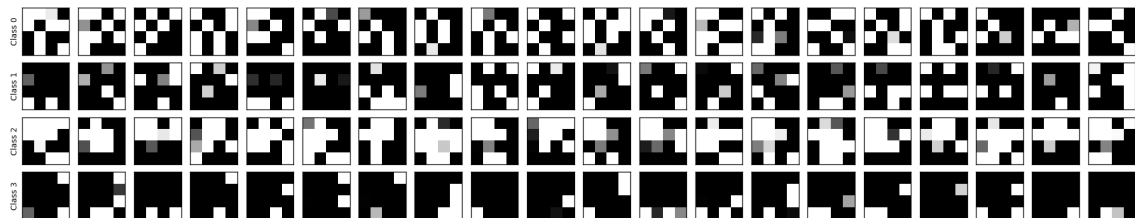


Figure: Obtained dataset

Shift type	Accuracy (%)
Original	98.78 ± 8.82
Shift horizontal (1px)	95.23 ± 10.12
Shift vertical (1px)	94.56 ± 11.34
Shift horizontal+vertical (1px)	89.12 ± 14.56

CNN Architecture: Conv2d with kernel 2×2 , stride 1 $\rightarrow$ ReLU $\rightarrow$ max pooling (2×2) $\rightarrow$ flatten $\rightarrow$ linear layer $\rightarrow$ class logits.

Meta-loss function: $\mathcal{L} = \mathcal{L}_{MSE} + \mathcal{R}_{dist}$

Table: Accuracy on original and shifted images

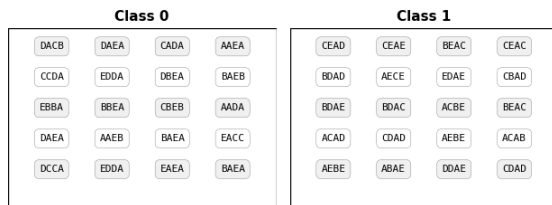


Figure: Obtained dataset

Generator: Trainable tensor.

RNN Architecture: RNN with hidden size 32, 2 layers → last hidden state → linear layer → class logits.

Parameters:

1. 2 classes, 20 samples per class
2. 10 steps in inner loop
3. Learning rates: inner = 0.5, outer = 0.005

Train data	Test: Original	Test: Reversed	Test: Permuted
Original	100.00 ± 0.00	0.00 ± 0.00	50.12 ± 17.01
Reversed	0.05 ± 0.79	99.80 ± 3.57	49.39 ± 17.25
Permuted	38.88 ± 22.07	37.27 ± 19.59	38.42 ± 14.98

Table: Accuracy on original, reversed and permuted test sets for models trained on different data types

Conclusion

- 1) We consider the problem of extracting the inductive bias of differentiable learning algorithms without prior knowledge of the input data distribution.
- 2) We propose a generative meta-learning framework where a generative model is trained to produce synthetic easy-to-generalize datasets that reflect the target model's inductive bias.
- 3) Experiments on CNNs and RNNs demonstrate that the generated datasets reflect the well-known inductive biases of these architectures: locality and translation invariance for CNNs, and sequential structure for RNNs.
- 4) Our method successfully extracts inductive bias without requiring any external data, offering a tool for analyzing different model architectures.